

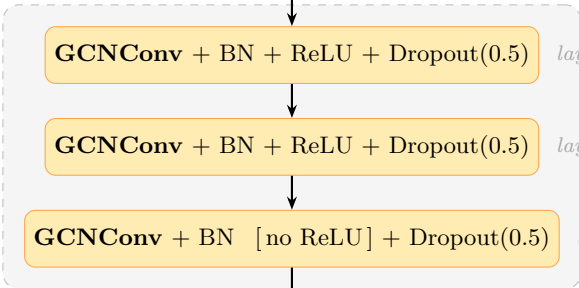
x: node type & attribute

node_depth

edge_index, edge_attr $\in R^{E \times 2}$

emb_dim=300

ASTNodeEncoder
type-emb + attr-emb + depth-emb



layer 1

layer 2

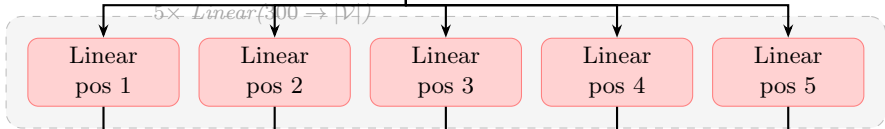
layer 3

$$\begin{aligned} msg &= \sigma(x_j + EdgeEnc(e_{ij})) \\ &\cdot D_{ii}^{-12} D_{jj}^{-12} \\ h_i &= agg + ReLU(x_i + r) deg_i \end{aligned}$$

R^{300}

Global Mean Pooling
 $h_g = \frac{1}{|V|} \sum_i h_i$

5x Linear(300 $\rightarrow$ |V|)



$\hat{y}_1$

$\hat{y}_2$

$\hat{y}_3$

$\hat{y}_4$

$\hat{y}_5$

Predicted code token sequence (F1 evaluated)